# My Solutions

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# 1 Problem 1

## 1.1 Coffee Shop Class Diagram

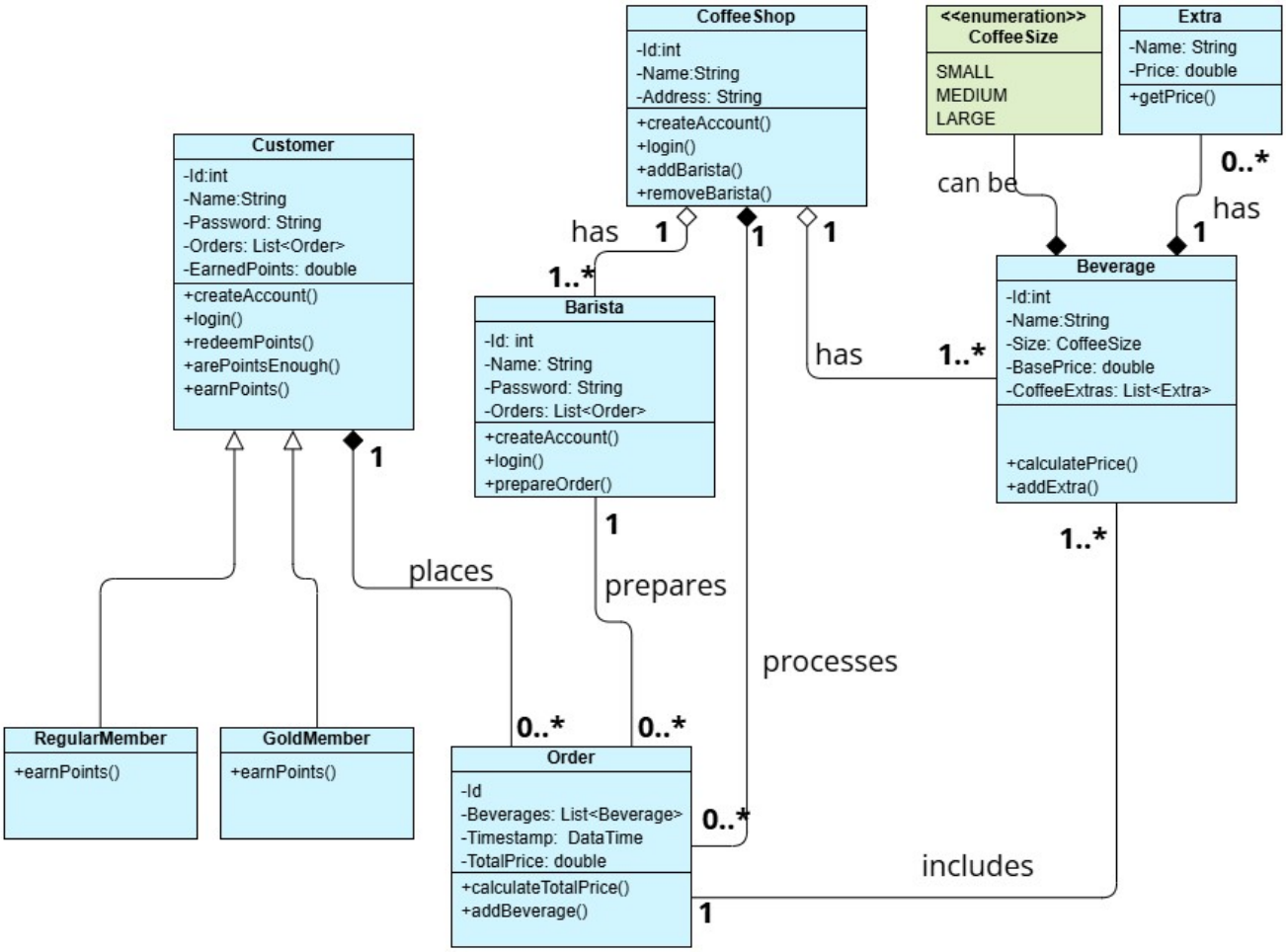


Figure 1: UML Class Diagram for Coffee Shop.

### 1.1.1 Explanation

- **Classes:** Coffee Shop, Customer(abstract), Barista, Beverage, Extra, RegularMember, GoldMember.

- **Enumerations:** CoffeeSize
- **Cardinalities:**
  1. Coffee shop and Customer: **many to many**.
  2. Customer can place 0 or more orders. There are two types of customers: Regular and Gold.
  3. Barista can prepare 0 or more orders.
  4. Order can have 1 or more beverages
  5. Beverage can have 0 or more extras and can be only SMALL, MEDIUM or LARGE.
- **Relationships:**
  - **Association:** barista and order, beverage and order.
  - **Aggregation:** coffee shop and barista, coffee shop and beverage.
  - **Composition:** beverage and Extra, beverage and CoffeSize, customer and order, coffee shop and order.
  - **Inheritance:** customer and regular member, customer and gold member.

## 1.2 Entity Relationship Diagram

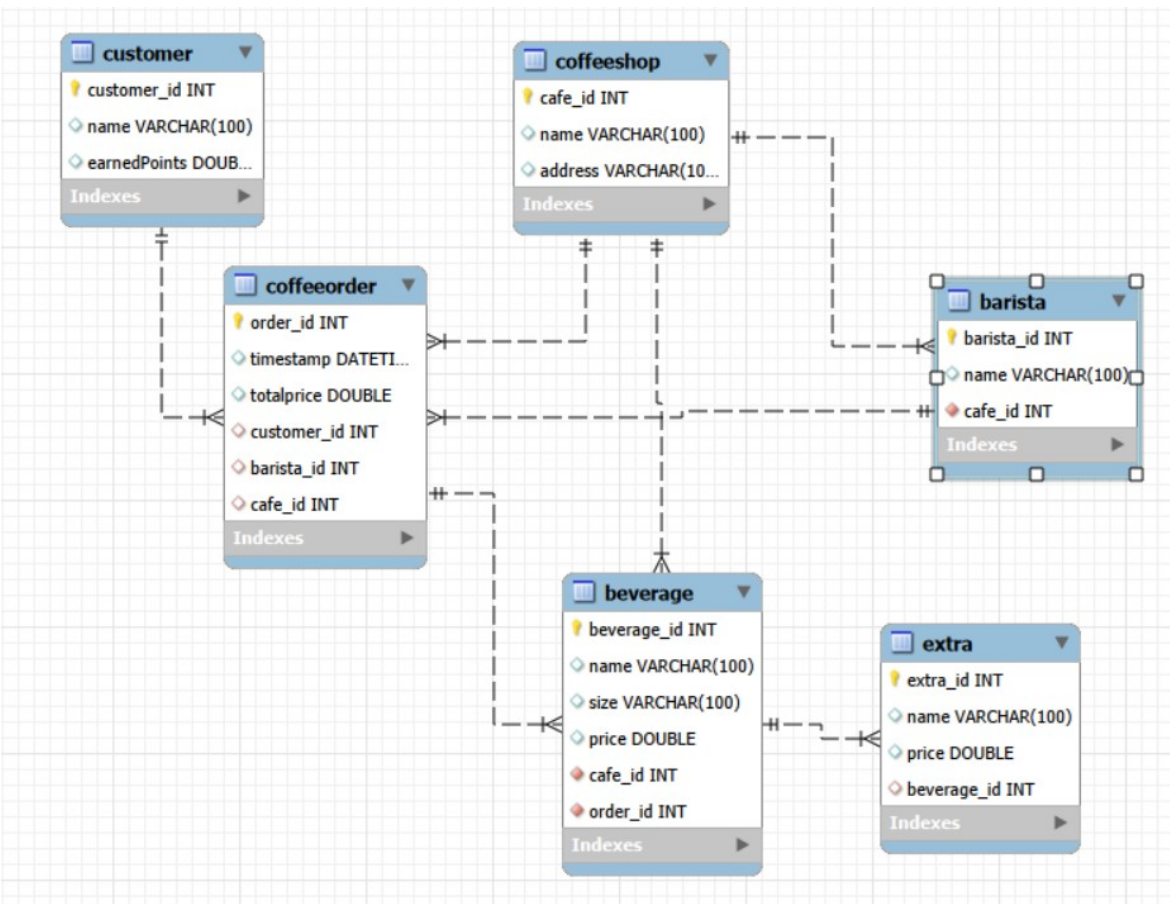


Figure 2: ER Diagram for Coffee Shop.

In total 6 tables. The Cardinalities:

- **Many to many:** coffeeshop and customer(**coffeeorder** serves as a join table to correctly express many to many relationship between customer and coffee shop).

- **One to many:** coffeeshop and barista(barista has a **cafe_id** as a foreign key); coffeeshop and beverage(beverage has a **cafe_id** as a foreign key); extra and beverage(extra has a **beverage_id** as a foreign key); order and beverage(beverage has an **order_id** as a foreign key);

# 2 Problem 2

Please follow this link to see the solution on git repo: SieMarket.